

Constraint Analysis

2.1 Concept

The design process starts by considering the forces that act on the aircraft, namely, lift, drag, thrust, and weight. This approach will lead to the fortunate discovery that several of the leading performance requirements of the Request for Proposal (RFP) can be translated into functional relationships between the minimum thrust-to-weight or thrust loading at sea-level takeoff (T_{SL}/W_{TO}) and wing loading at takeoff (W_{TO}/S). The keys to the development of these relationships, and a typical step in any design process, are reasonable assumptions for the aircraft lift-drag polar and the lapse of the engine thrust with flight altitude and Mach number. It is not necessary that these assumptions be exact, but greater accuracy reduces the need for iteration. It is possible to satisfy these aircraft/engine system requirements as long as the thrust loading at least equals the largest value found at the selected wing loading. Notice that the more detailed aspects of design, such as stability, control, configuration layout, and structures, are set aside for later consideration by aircraft system designers.

An example of the results of a typical constraint analysis is portrayed in Fig. 2.1. Shown there are the minimum T_{SL}/W_{TO} as a function of W_{TO}/S needed for the following: 1) takeoff from a runway of given length; 2) flight at a given altitude and required speed; 3) turn at a given altitude, speed, and required rate; and 4) landing without reverse thrust on a runway of given length. Any of the trends that are not familiar will be made clear by the analysis of the next section.

What is important to realize about Fig. 2.1 is that any combination of T_{SL}/W_{TO} and W_{TO}/S that falls in the “solution space” shown there automatically meets all of the constraints considered. For better or for worse, there are many acceptable solutions available at this point. It is important to identify which is “best” and why.

It is possible to include many other performance constraints, such as required service ceiling and acceleration time, on the same diagram. By incorporating all known constraints, the range of acceptable loading parameters (that is, the solution space) will be appropriately restricted.

A look at example records of thrust loading vs wing loading at takeoff is quite interesting. Figures 2.2 and 2.3 represent collections of the design points for jet engine powered transport-type and fighter-type aircraft, respectively. The thing that leaps out of these figures is the diversity of design points. The selected design point is very sensitive to the application and the preferences of the designer. Pick out some of your favorite airplanes and see if you can explain their location on the constraint diagram. For example, the low wing loadings of the C-20A and C-21A are probably caused by short takeoff length requirements, whereas the high thrust loading and low wing loading of the YF-22 and MIG-31 are probably caused by requirements for specific combat performance in both the subsonic and supersonic

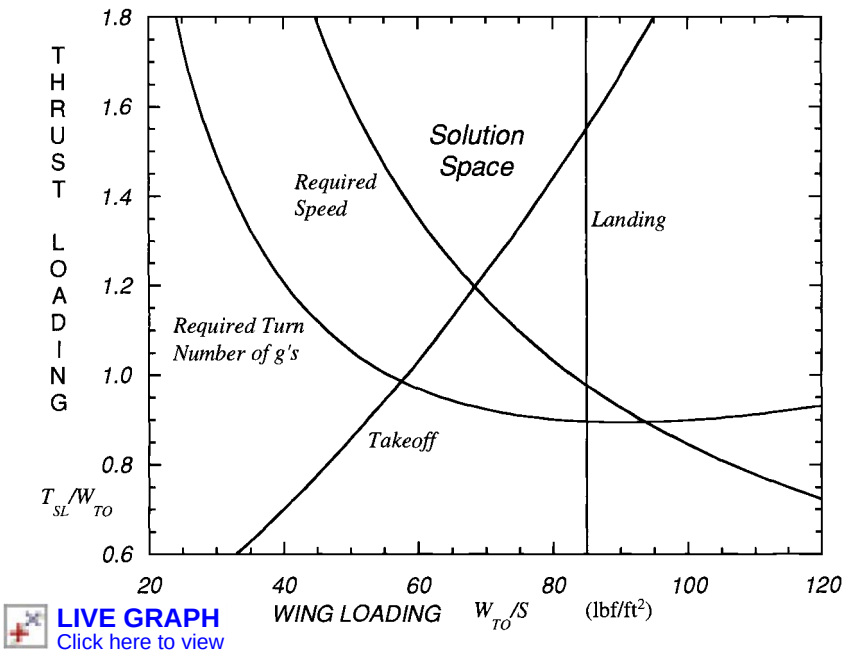


Fig. 2.1 Constraint analysis—thrust loading vs wing loading.

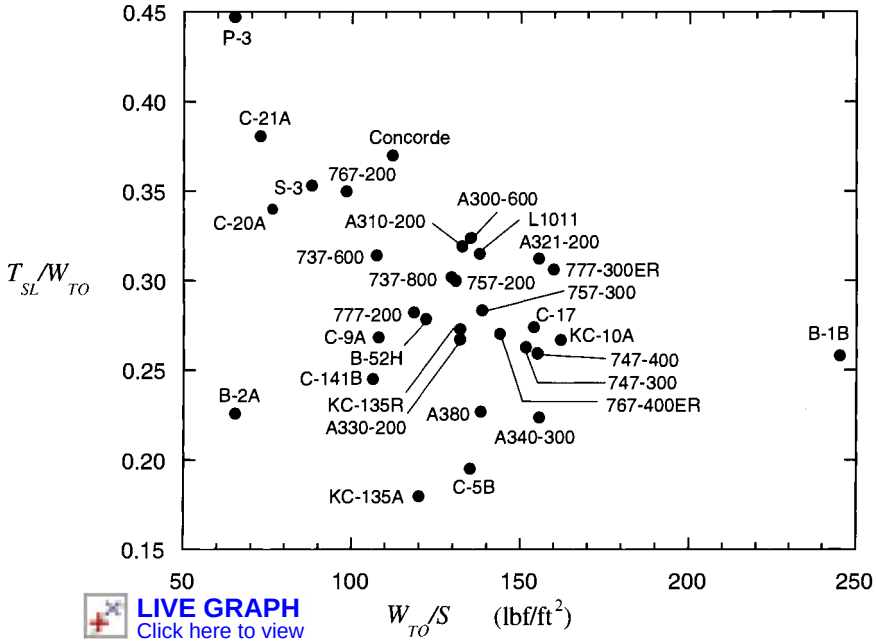


Fig. 2.2 Thrust loading vs wing loading—cargo and passenger aircraft.^{1,2}

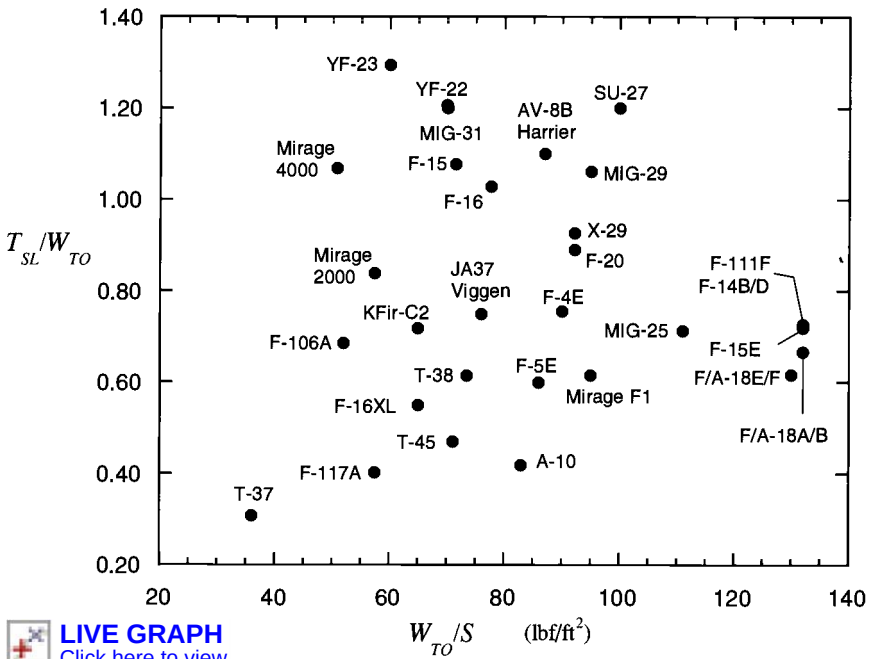


Fig. 2.3 Thrust loading vs wing loading—fighter aircraft.^{1,2}

arenas. Where do you think the location would be for the AAF, supersonic business jet, GRA, and UAV?

2.2 Design Tools

A “master equation” for the flight performance of aircraft in terms of takeoff thrust loading (T_{SL}/W_{TO}) and wing loading (W_{TO}/S) can be derived directly from force considerations. We treat the aircraft, shown in Fig. 2.4, as a point mass with a velocity (V) in still air at a flight path angle of θ to the horizon. The velocity of the air ($-V$) has an angle of attack (AOA) to the wing chord line (WCL). The lift

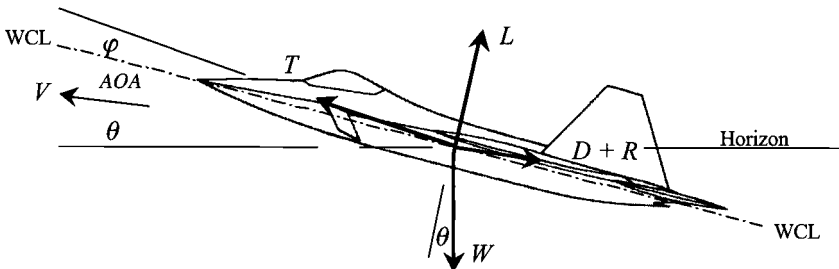


Fig. 2.4 Forces on aircraft.

(L) and drag ($D + R$) forces are normal and parallel to this velocity, respectively. The thrust (T) is at an angle φ to the wing chord line (usually small). Applying Newton's second law to this aircraft we have the following for the accelerations:

Parallel to V :

$$T \cos(AOA + \varphi) - W \sin \theta - (D + R) = \frac{W}{g_0} a_{\parallel} = \frac{W}{g_0} \frac{dV}{dt} \quad (2.1a)$$

Perpendicular to V :

$$L + T \sin(AOA + \varphi) - W \cos \theta = \frac{W}{g_0} a_{\perp} \quad (2.1b)$$

Multiplying Eq. (2.1a) by the velocity (V), we have the following equation in the direction of flight:

$$\{T \cos(AOA + \varphi) - (D + R)\}V = W \left\{ V \sin \theta + \frac{d}{dt} \left(\frac{V^2}{2g_0} \right) \right\} \quad (i)$$

Note that for most flight conditions the thrust is very nearly aligned with the direction of flight, so that the angle ($AOA + \varphi$) is small and thus $\cos(AOA + \varphi) \approx 1$. This term will therefore be dropped from the ensuing development, but it could be restored if necessary or desirable. Also, multiplying by velocity has transformed a force relationship into a power, or time rate of change of energy, equation. This will have profound consequences in what follows. Since $V \sin \theta$ is simply the time rate of change of altitude (h) or

$$V \sin \theta = \frac{dh}{dt} \quad (ii)$$

then combining equations (i) and (ii) and dividing by W gives

$$\frac{T - (D + R)}{W} V = \frac{d}{dt} \left\{ h + \frac{V^2}{2g_0} \right\} = \frac{dz_e}{dt} \quad (2.2a)$$

where $z_e = h + V^2/2g_0$ represents the sum of instantaneous potential and kinetic energies of the aircraft and is frequently referred to as the "energy height." The energy height may be most easily visualized as the altitude the aircraft would attain if its kinetic energy were completely converted into potential energy. Lines of constant energy height are plotted in Fig. 2.5 vs the altitude-velocity axes. The flight condition ($h = 20$ kft and $V = 1134$ fps) marked with a star in Fig. 2.5 corresponds to an energy height of 40 kft. Excess power is required to increase the energy height of the aircraft, and the rate of change is proportional to the amount of excess power.

The left-hand side of Eq. (2.2a) in modern times has become recognized as a dominant property of the aircraft and is called the weight specific excess power

$$P_s = \frac{dz_e}{dt} = \frac{d}{dt} \left\{ h + \frac{V^2}{2g} \right\} \quad (2.2b)$$

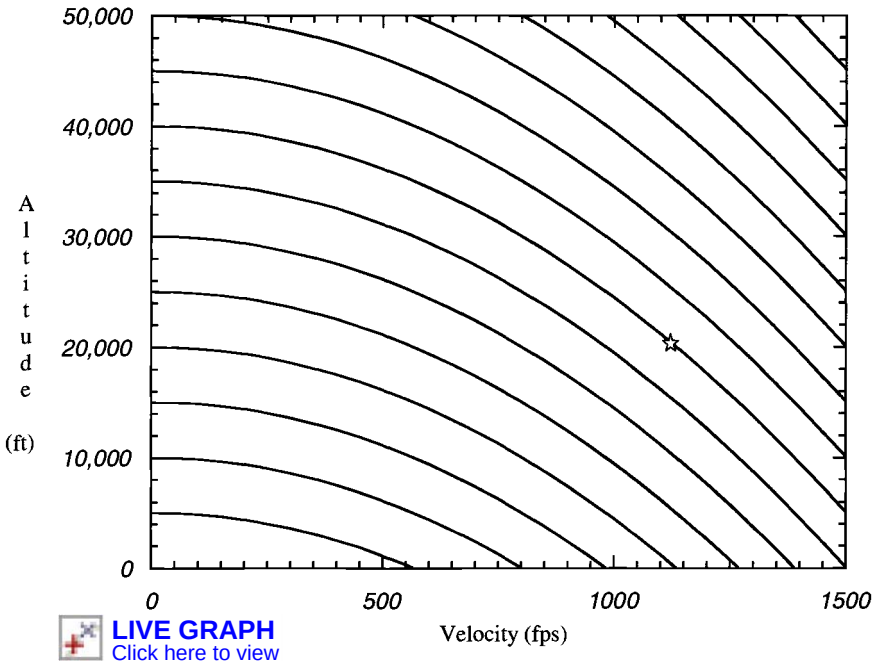


Fig. 2.5 Lines of constant energy height (z_e).

This is a powerful grouping for understanding and predicting the dynamics of flight, including both rate of climb (dh/dt) and acceleration (dV/dt) capabilities. P_s must have the units of velocity.

If we assume that the installed thrust is given by

$$T = \alpha T_{SL} \quad (2.3)$$

where α is the installed full throttle thrust lapse, which depends on altitude, speed, and whether or not an afterburner is operating, and the instantaneous weight is given by

$$W = \beta W_{TO} \quad (2.4)$$

where β depends on how much fuel has been consumed and payload delivered, then Eq. (2.2a) becomes

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left(\frac{D + R}{\beta W_{TO}} + \frac{P_s}{V} \right) \quad (2.5)$$

Note: Definition of α

Particular caution must be exercised in the use of α throughout this textbook because it is intended to be referenced only to the *maximum* thrust or power available for the prevailing engine configuration and flight condition. For example, an afterburning engine will have two possible values of α for any flight condition,

one for mil power and one for max power (see Sec. 1.11E of RFP). You must differentiate carefully between them. Lower values of thrust are always available by simply throttling the fuel flow, thrust, or power.

It is equally important to remember that both T and T_{SL} refer to the “installed” engine thrust, which is generally less than the “uninstalled” engine thrust that would be produced if the external flow were ideal and created no drag. The difference between them is the additional drag generated on the external surfaces, which is strongly influenced by the presence of the engine and is not included in the aircraft drag model. The additional drag is usually confined to the inlet and exhaust nozzle surfaces, but in unfavorable circumstances can be found anywhere, including adjacent fuselage, wing, and tail surfaces. The subject of “installed” vs “uninstalled” thrust is dealt with in detail in Chapter 6 and Appendix E.

Now, using the traditional aircraft lift and drag relationships,

$$L = nW = qC_L S \quad (2.6)$$

where n = load factor = number of g 's ($g = g_0$) $\perp$ to V ($n = 1$ for straight and level flight even when $dV/dt \neq 0$),

$$D = qC_D S \quad (2.7a)$$

and

$$R = qC_{DR} S \quad (2.7b)$$

where D and C_D refer to the “clean” or basic aircraft and R and C_{DR} refer to the additional drag caused, for example, by external stores, braking parachutes or flaps, or temporary external hardware. Then

$$C_L = \frac{nW}{qS} = \frac{n\beta}{q} \left(\frac{W_{TO}}{S} \right) \quad (2.8)$$

Further, assuming the lift-drag polar relationship,

$$C_D = K_1 C_L^2 + K_2 C_L + C_{D0} \quad (2.9)$$

Equations (2.7–2.9) can be combined to yield

$$D + R = qS \left\{ K_1 \left(\frac{n\beta}{q} \frac{W_{TO}}{S} \right)^2 + K_2 \left(\frac{n\beta}{q} \frac{W_{TO}}{S} \right) + C_{D0} + C_{DR} \right\} \quad (2.10)$$

Finally, Eq. (2.10) may be substituted into Eq. (2.5) to produce the general form of the “master equation”

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ \frac{qS}{\beta W_{TO}} \left[K_1 \left(\frac{n\beta}{q} \frac{W_{TO}}{S} \right)^2 + K_2 \left(\frac{n\beta}{q} \frac{W_{TO}}{S} \right) + C_{D0} + C_{DR} \right] + \frac{P_s}{V} \right\} \quad (2.11)$$

It should be clear that Eq. (2.11) will provide the desired relationships between T_{SL}/W_{TO} and W_{TO}/S that become constraint diagram boundaries. It should also be evident that the general form of Eq. (2.11) is such that there is one value of

W_{TO}/S for which T_{SL}/W_{TO} is minimized, as seen in Fig. 2.1. This important fact will be elaborated upon in the example cases that follow.

Note: Lift-drag polar equation

The conventional form of the lift-drag polar equation is³

$$C_D = C_{D\min} + K' C_L^2 + K''(C_L - C_{L\min})^2$$

where K' is the inviscid drag due to lift (induced drag) and K'' is the viscous drag due to lift (skin friction and pressure drag). Expanding and collecting like terms shows that the lift-drag polar equation may also be written

$$C_D = (K' + K'')C_L^2 - (2K''C_{L\min})C_L + (C_{D\min} + K''C_{L\min}^2)$$

or

$$C_D = K_1 C_L^2 + K_2 C_L + C_{D0} \quad (2.9)$$

where

$$K_1 = K' + K''$$

$$K_2 = -2K''C_{L\min}$$

$$C_{D0} = C_{D\min} + K''C_{L\min}^2$$

Note that the physical interpretation of C_{D0} is the drag coefficient at zero lift. Also, for most high-performance aircraft $C_{L\min} \approx 0$, so that $K_2 \approx 0$.

A large number and variety of special cases of Eq. (2.11) will be developed in order both to illustrate its behavior and to provide more specific design tools for constraint analysis. In all of the example cases that follow, it is assumed that the α of Eq. (2.3), the β of Eq. (2.4), and the K_1 , K_2 , and C_{D0} of Eq. (2.9) are known. If they change significantly over the period of flight being analyzed, either piecewise solution or use of representative working averages should be considered.

2.2.1 Case 1: Constant Altitude/Speed Cruise ($P_s = 0$)

Given: $dh/dt = 0$, $dV/dt = 0$, $n = 1$ ($L = W$), and values of h and V (i.e., q). Under these conditions Eq. (2.11) becomes

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + K_2 + \frac{C_{D0} + C_{DR}}{\beta/q(W_{TO}/S)} \right\} \quad (2.12)$$

This relationship is quite complex because T_{SL}/W_{TO} grows indefinitely large as W_{TO}/S becomes very large or very small. The location of the minimum for T_{SL}/W_{TO} can be found by differentiating Eq. (2.12) with respect to W_{TO}/S and setting the result equal to zero. This leads to

$$\left[\frac{W_{TO}}{S} \right]_{\min T/W} = \frac{q}{\beta} \sqrt{\frac{C_{D0} + C_{DR}}{K_1}}$$

and

$$\left[\frac{T_{SL}}{W_{TO}} \right]_{\min} = \frac{\beta}{\alpha} \{ 2\sqrt{(C_{D0} + C_{DR})K_1} + K_2 \}$$

This condition produces the minimum thrust and drag for the aircraft and generally corresponds to maximum range.

Also of special note is the frequently encountered situation where the required q is so large that

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \frac{C_{D0} + C_{DR}}{\beta/q(W_{TO}/S)}$$

or

$$\left(\frac{T_{SL}}{W_{TO}} \right) \left(\frac{W_{TO}}{S} \right) = \frac{q(C_{D0} + C_{DR})}{\alpha} \quad (2.13)$$

This limiting case, which corresponds to $C_D = C_{D0} + C_{DR}$, therefore displays a hyperbolic relationship between T_{SL}/W_{TO} and W_{TO}/S .

2.2.2 Case 2: Constant Speed Climb ($P_s = dh/dt$)

Given: $dV/dt = 0$, $n \approx 1$, ($L \approx W$), and values of h , $dh/dt > 0$, and V (i.e., q). Under these conditions Eq. (2.11) becomes

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + K_2 + \frac{C_{D0} + C_{DR}}{\beta/q(W_{TO}/S)} + \frac{1}{V} \frac{dh}{dt} \right\} \quad (2.14)$$

Since $(1/V)(dh/dt)$ is a constant, the behavior of this equation is not only the same as that of Eq. (2.12), but the minimum value of T_{SL}/W_{TO} occurs again at

$$\left[\frac{W_{TO}}{S} \right]_{\min T/W} = \frac{q}{\beta} \sqrt{\frac{C_{D0} + C_{DR}}{K_1}}$$

and

$$\left[\frac{T_{SL}}{W_{TO}} \right]_{\min} = \frac{\beta}{\alpha} \left\{ 2\sqrt{(C_{D0} + C_{DR})K_1} + K_2 + \frac{1}{V} \frac{dh}{dt} \right\}$$

2.2.3 Case 3: Constant Altitude/Speed Turn ($P_s = 0$)

Given: $dh/dt = 0$, $dV/dt = 0$, and values of h , V (i.e., q), and $n > 1$. Under these conditions Eq. (2.11) becomes

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 n^2 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + K_2 n + \frac{C_{D0} + C_{DR}}{\beta/q(W_{TO}/S)} \right\} \quad (2.15)$$

Again, the behavior of this equation is the same as that of Eq. (2.12), the minimum of T_{SL}/W_{TO} occurring at

$$\left[\frac{W_{TO}}{S} \right]_{\min T/W} = \frac{q}{n\beta} \sqrt{\frac{C_{D0} + C_{DR}}{K_1}}$$

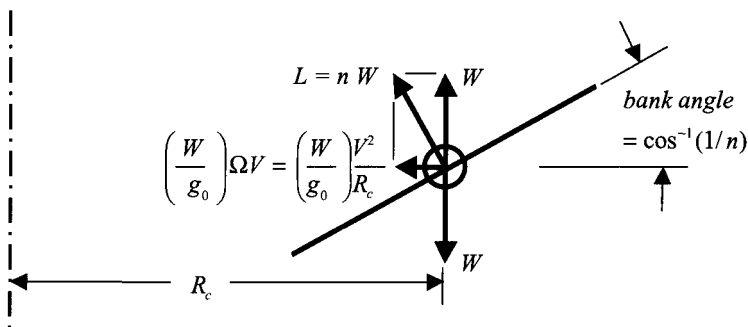


Fig. 2.6 Forces on aircraft in turn.

and

$$\left[\frac{T_{SL}}{W_{TO}} \right]_{\min} = \frac{n\beta}{\alpha} \{ 2\sqrt{(C_{D0} + C_{DR})K_1} + K_2 \}$$

Occasionally n is stipulated in terms of other quantities. In the case of a level, constant velocity turn, for example, where the vertical component of lift balances the weight and the horizontal component is the centripetal force, n is as shown in Fig. 2.6.

It follows from the Pythagorean Theorem that

$$n = \sqrt{1 + \left(\frac{\Omega V}{g_0} \right)^2} \quad (2.16)$$

and

$$n = \sqrt{1 + \left(\frac{V^2}{g_0 R_c} \right)^2} \quad (2.17)$$

which can be used when the rotation rate (Ω) or the radius of curvature (R_c) is given rather than the load factor (n).

2.2.4 Case 4: Horizontal Acceleration [$P_s = (V/g_0) (dV/dt)$]

Given: $dh/dt = 0$, $n = 1$ ($L = W$) and values of h , $V_{initial}$, V_{final} , and $\Delta t_{allowable}$. Under these conditions Eq. (2.11) becomes

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + K_2 + \frac{C_{D0} + C_{DR}}{\beta/q(W_{TO}/S)} + \frac{1}{g_0} \frac{dV}{dt} \right\} \quad (2.18a)$$

which can be rearranged to yield

$$\frac{1}{g_0} \frac{dV}{dt} = \frac{\alpha}{\beta} \frac{T_{SL}}{W_{TO}} - \left\{ K_1 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + K_2 + \frac{C_{D0} + C_{DR}}{\beta/q(W_{TO}/S)} \right\} \quad (2.18b)$$

Strictly speaking, Eq. (2.18b) must be integrated from $V_{initial}$ to V_{final} in order to find combinations of thrust loading and wing loading that satisfy the acceleration

time criterion. A useful approximation, however, is to select some point between $V_{initial}$ and V_{final} at which the quantities in Eq. (2.18a) approximate their working average over that range, and set

$$\frac{1}{g_0} \frac{dV}{dt} = \frac{1}{g_0} \left(\frac{V_{final} - V_{initial}}{\Delta t_{allowable}} \right)$$

In this case the constraint curve is obtained from Eq. (2.18a), which again has the properties of Eq. (2.12) including the location of the minimum T_{SL}/W_{TO} .

Equation (2.18b) can be numerically integrated by first recasting the equation as

$$\Delta t_{allowable} = \frac{1}{g_0} \int_{V_{initial}}^{V_{final}} \frac{V dV}{P_s} = \frac{1}{2g_0} \int_{V_{initial}^2}^{V_{final}^2} \frac{dV^2}{P_s} \quad (2.19a)$$

where

$$P_s = V \left\{ \frac{\alpha}{\beta} \frac{T_{SL}}{W_{TO}} - \left[K_1 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + K_2 + \frac{C_{D0} + C_{DR}}{\beta/q(W_{TO}/S)} \right] \right\} \quad (2.19b)$$

The solution of the required thrust loading (T_{SL}/W_{TO}) for each wing loading (W_{TO}/S) can be obtained by the following procedure:

1) Divide the change in kinetic energy into even sized increments, and calculate the minimum thrust loading [maximum T_{SL}/W_{TO} corresponding to $P_s = 0$ in Eq. (2.19b) for all kinetic energy states].

2) Select a thrust loading larger than the minimum and calculate the acceleration time (Δt) using Eqs. (2.19a) and (2.19b). Compare resulting acceleration time with $\Delta t_{allowable}$. Change the thrust loading and recalculate the acceleration time until it matches $\Delta t_{allowable}$.

2.2.5 Case 5: Takeoff Ground Roll (s_G), when $T_{SL} \gg (D + R)$

Given: $dh/dt = 0$ and values of s_G , ρ , C_{Lmax} , and $V_{TO} = k_{TO} V_{STALL}$.

Under these conditions Eq. (2.5) reduces to

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha g_0} \frac{dV}{dt} = \frac{\beta}{\alpha g_0} \frac{dV}{ds/V}$$

which can be rearranged to yield

$$ds = \frac{\beta}{\alpha g_0} \left(\frac{W_{TO}}{T_{SL}} \right) V dV$$

and integrated from $s = 0$ and $V = 0$ to takeoff, where $s = s_G$ and $V = V_{TO}$, with the result that

$$s_G = \frac{\beta}{\alpha} \left(\frac{W_{TO}}{T_{SL}} \right) \frac{V_{TO}^2}{2g_0}$$

provided that representative takeoff values of α and β are used. Defining

$$V_{TO} = k_{TO} V_{STALL} \quad (2.20)$$

where k_{TO} is a constant greater than one (generally specified by appropriate flying regulations) and V_{STALL} is the minimum speed at which the airplane flies at $C_{L \max}$, then

$$q C_{L \max} S = \frac{1}{2} \rho V_{STALL}^2 C_{L \max} S = \beta W_{TO}$$

or

$$\frac{V_{TO}^2}{2} = k_{TO}^2 \frac{V_{STALL}^2}{2} = \frac{\beta k_{TO}^2}{\rho C_{L \max}} \left(\frac{W_{TO}}{S} \right) \quad (2.21)$$

with the final result that

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta^2}{\alpha} \frac{k_{TO}^2}{s_G \rho g_0 C_{L \max}} \left(\frac{W_{TO}}{S} \right) \quad (2.22)$$

For this limiting case T_{SL}/W_{TO} is directly proportional to W_{TO}/S and inversely proportional to s_G .

2.2.6 Case 6: Takeoff Ground Roll (s_G)

Given: $dh/dt = 0$ and values of ρ , $D = q C_D S$, $C_{L \max}$, $V_{TO} = k_{TO} V_{STALL}$, and $R = q C_{DR} S + \mu_{TO}(\beta W_{TO} - q C_L S)$.

Under these conditions

$$\frac{D + R}{\beta W_{TO}} = \frac{(C_D + C_{DR} - \mu_{TO} C_L) q S + \mu_{TO} \beta W_{TO}}{\beta W_{TO}}$$

so that Eq. (2.5) becomes

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ \xi_{TO} \frac{q}{\beta} \left(\frac{S}{W_{TO}} \right) + \mu_{TO} + \frac{1}{g_0} \frac{dV}{dt} \right\} \quad (2.23)$$

where

$$\xi_{TO} = C_D + C_{DR} - \mu_{TO} C_L \quad (2.24)$$

which can be rearranged and integrated, as in Case 5, to yield

$$s_G = -\frac{\beta(W_{TO}/S)}{\rho g_0 \xi_{TO}} \ln \left\{ 1 - \xi_{TO} / \left[\left(\frac{\alpha}{\beta} \frac{T_{SL}}{W_{TO}} - \mu_{TO} \right) \frac{C_{L \max}}{k_{TO}^2} \right] \right\} \quad (2.25)$$

provided that representative takeoff values of α and β are used.

For this case T_{SL}/W_{TO} must increase continuously with W_{TO}/S , but in a more complex manner than in Case 5. That result is again obtained in the limit as all terms in ξ_{TO} approach zero and $\ln(1 - \varepsilon)$ approaches $-\varepsilon$, whence

$$s_G \rightarrow \frac{\beta(W_{TO}/S)}{\rho g_0 \xi_{TO}} \frac{\xi_{TO}}{(\alpha/\beta)(T_{SL}/W_{TO}) C_{L \max} / k_{TO}^2}$$

or

$$\frac{T_{SL}}{W_{TO}} \rightarrow \frac{\beta^2}{\alpha} \frac{k_{TO}^2}{s_G \rho g_0 C_{L \max}} \left(\frac{W_{TO}}{S} \right) \quad (2.22)$$

a)

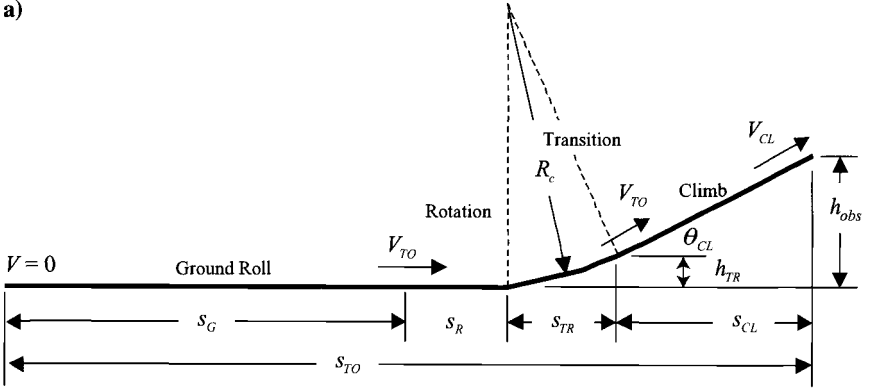


Fig. 2.7a Takeoff terminology ($h_{TR} < h_{obs}$).

Note: Total takeoff distance

Two cases arise:

Case A (Fig. 2.7a): The total takeoff distance (s_{TO}) can be analyzed as ground roll (s_G) plus three other distances: the first (s_R) to rotate the aircraft to the takeoff lift condition (traditionally $C_L = 0.8C_{L\max}$) while still on the ground; the second (s_{TR}) to transit to the angle of climb direction; and the last (s_{CL}) to clear an obstacle of given height. These distances may be estimated as follows¹:

$$s_R = t_R V_{TO} = t_R k_{TO} \sqrt{\{2\beta/(\rho C_{L\max})\}(W_{TO}/S)} \quad (2.26)$$

where t_R is a total aircraft rotation time based on experience (normally 3 s),

$$s_{TR} = R_c \sin \theta_{CL} = \frac{V_{TO}^2 \sin \theta_{CL}}{g_0(n-1)}$$

$$s_{TR} = \frac{V_{TO}^2 \sin \theta_{CL}}{g_0(0.8k_{TO}^2 - 1)} = \frac{k_{TO}^2 \sin \theta_{CL}}{g_0(0.8k_{TO}^2 - 1)} \frac{2\beta}{\rho C_{L\max}} \left(\frac{W_{TO}}{S} \right) \quad (2.27)$$

where θ_{CL} is the angle of climb, which can in turn be obtained from Eq. (2.2a) as

$$\frac{1}{V} \frac{dh}{dt} = \sin \theta_{CL} = \frac{T - D}{W}$$

and, if $h_{obs} > h_{TR}$

$$s_{CL} = \frac{h_{obs} - h_{TR}}{\tan \theta_{CL}} \quad (2.28)$$

where h_{obs} is the required clearance height and h_{TR} is given by the following expression, provided that $h_{obs} > h_{TR}$:

$$h_{TR} = \frac{V_{TO}^2(1 - \cos \theta_{CL})}{g_0(0.8k_{TO}^2 - 1)} = \frac{k_{TO}^2(1 - \cos \theta_{CL})}{g_0(0.8k_{TO}^2 - 1)} \frac{2\beta}{\rho C_{L\max}} \left(\frac{W_{TO}}{S} \right) \quad (2.29)$$

b)

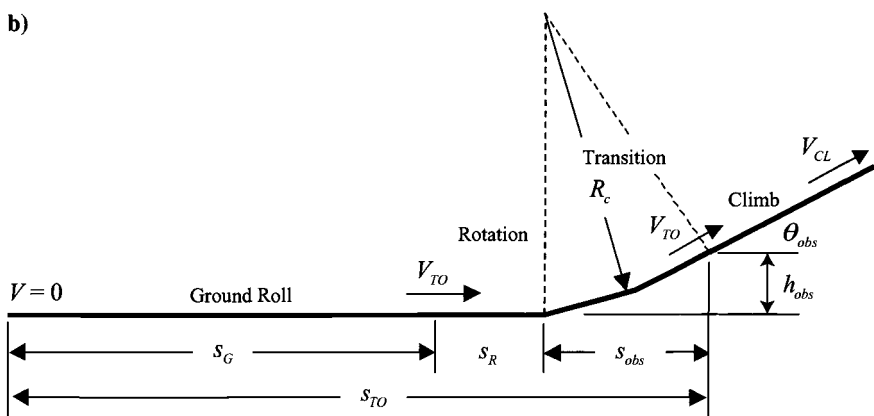


Fig. 2.7b Takeoff terminology ($h_{TR} > h_{obs}$).

Case B (Fig. 2.7b): The distances s_G and s_R are the same as in case A, but the obstacle is cleared during transition so $s_{TO} = s_G + s_R + s_{obs}$ where s_{obs} is the distance from the end of rotation to the point where the height h_{obs} is attained. There follows

$$s_{obs} = R_c \sin \theta_{obs} = \frac{V_{TO}^2 \sin \theta_{obs}}{g_0(0.8k_{TO}^2 - 1)} \quad (2.30)$$

where

$$\theta_{obs} = \cos^{-1} \left(1 - \frac{h_{obs}}{R_c} \right)$$

2.2.7 Case 7: Braking Roll (s_B)

Given: $\alpha \leq 0$ (reverse thrust), $dh/dt = 0$ and values of ρ , $V_{TD} = k_{TD} V_{STALL}$, $D = q C_D S$, and $R = q C_{DR} S + \mu_B (\beta W_{TO} - q C_L S)$.

Under these conditions

$$\frac{D + R}{\beta W_{TO}} = \frac{(C_D + C_{DR} - \mu_B C_L) q + \mu_B \beta W_{TO}}{\beta W_{TO}}$$

so that Eq. (2.5) becomes

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ \xi_L \frac{q}{\beta} \left(\frac{S}{W_{TO}} \right) + \mu_B + \frac{1}{g_0} \frac{dV}{dt} \right\} \quad (2.31)$$

where

$$\xi_L = C_D + C_{DR} - \mu_B C_L \quad (2.32)$$

which can be rearranged and integrated, as in Case 6, to yield

$$s_G = \frac{\beta(W_{TO}/S)}{\rho g_0 \xi_L} \ln \left\{ 1 + \xi_L \left/ \left[\left(\mu_B + \frac{(-\alpha)}{\beta} \frac{T_{SL}}{W_{TO}} \right) \frac{C_{Lmax}}{k_{TD}^2} \right] \right. \right\} \quad (2.33)$$

provided that representative landing values of α and β are used and where k_{TD} is a constant greater than one (generally specified by appropriate flying regulations).

For this case reverse thrust can be used to great advantage. If the second term in the bracket of Eq. (2.33) is made much less than one by making $(-\alpha)$ very large, then

$$s_B \rightarrow \frac{\beta(W_{TO}/S)}{\rho g_0 \xi_L} \frac{\xi_L}{[(-\alpha)/\beta](T_{SL}/W_{TO})(C_{L\max}/k_{TD}^2)}$$

or

$$\frac{T_{SL}}{W_{TO}} \rightarrow \frac{\beta^2}{(-\alpha)} \frac{k_{TD}^2}{s_B \rho g_0 C_{L\max}} \left(\frac{W_{TO}}{S} \right) \quad (2.34)$$

Note: Total landing distance

The total landing distance (s_L) can be analyzed as braking roll (s_B) plus two other distances (see Fig. 2.8): the first (s_A) to clear an obstacle of given height and the second (s_{FR}) a free roll traversed before the brakes are fully applied. These distances may be estimated as follows:¹

$$s_A = \frac{2\beta}{\rho g_0 (C_D + C_{DR})} \left(\frac{W_{TO}}{S} \right) \left(\frac{k_{obs}^2 - k_{TD}^2}{k_{obs}^2 + k_{TD}^2} \right) + \frac{C_{L\max}}{(C_D + C_{DR})} \frac{2h_{obs}}{(k_{obs}^2 + k_{TD}^2)} \quad (2.35)$$

where h_{obs} is the height of the obstacle and the velocity at the obstacle is

$$V_{obs} = k_{obs} V_{STALL} \quad (2.36)$$

and

$$s_{FR} = t_{FR} V_{TD} = t_{FR} k_{TD} \sqrt{2\beta/(\rho C_{L\max})} (W_{TO}/S) \quad (2.37)$$

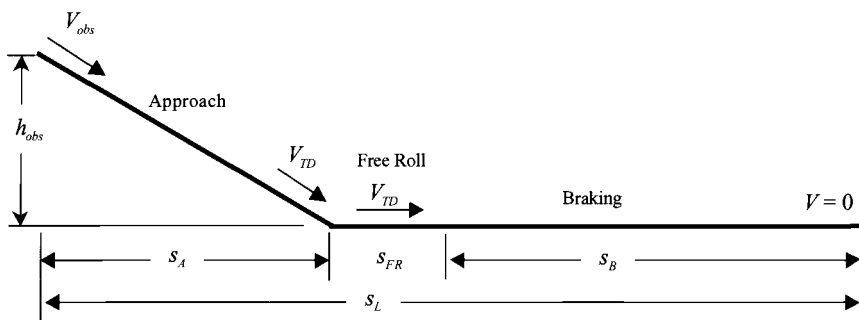


Fig. 2.8 Landing terminology.

where t_{FR} is a total system reaction time based on experience (normally 3 s) that allows for the deployment of a parachute or thrust reverser.

2.2.8 Case 8: Service Ceiling ($P_s = dh/dt$)

Given: $dV/dt = 0$, $n = 1$ ($L = W$), and the values of h (i.e., α), $dh/dt > 0$, and C_L .

Under these conditions Eq. (2.11) becomes

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + K_2 + \frac{C_{D0} + C_{DR}}{\beta/q(W_{TO}/S)} + \frac{1}{V} \frac{dh}{dt} \right\} \quad (2.38)$$

where

$$q C_L S = \beta W_{TO}$$

or

$$C_L = \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) \quad (2.39)$$

and

$$q = \frac{\rho V^2}{2} = \frac{\beta}{C_L} \left(\frac{W_{TO}}{S} \right)$$

or

$$V = \sqrt{\frac{2\beta}{\sigma \rho_{SL} C_L} \left(\frac{W_{TO}}{S} \right)} \quad (2.40)$$

so that

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 C_L + K_2 + \frac{C_{D0} + C_{DR}}{C_L} + \frac{1}{V} \frac{dh}{dt} \right\} \quad (2.41)$$

2.2.9 Case 9: Takeoff Climb Angle

Given: θ , $n = 1$ ($L = W$), $dV/dt = 0$, C_{DR} , $C_{L \max}$, k_{TO} , and the values of h and σ .

Under these conditions Eq. (2.11) becomes

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 \left(\frac{\beta W_{TO}}{q S} \right) + K_2 + \frac{C_{D0} + C_{DR}}{(\beta W_{TO}/q S)} + \sin \theta \right\} \quad (2.42)$$

Since

$$C_L = \frac{C_{L \max}}{k_{TO}^2} = \frac{W}{q S} = \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right)$$

then

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 \frac{C_{L \max}}{k_{TO}^2} + K_2 + \frac{C_{D0} + C_{DR}}{C_{L \max}/k_{TO}^2} + \sin \theta \right\} \quad (2.43)$$

and

$$V = V_{TO} = \sqrt{\frac{2\beta k_{TO}^2}{\sigma \rho_{SL} C_{L\max}}} \left(\frac{W_{TO}}{S} \right) \quad (2.44)$$

is employed to find M_{TO} for a given W_{TO}/S and thus the applicable values of α , K_1 , K_2 , and C_{D0} . Because they vary slowly with W_{TO}/S , the constraint boundary is a line of almost constant T_{SL}/W_{TO} .

2.2.10 Case 10: Carrier Takeoff

Given: $n = 1$ ($L = W$), V_{TO} , dV/dt , $C_{L\max}$, k_{TO} , β , and the values of h and σ . Solving Eq. (2.44) for wing loading gives

$$\left[\frac{W_{TO}}{S} \right]_{\max} = \frac{\sigma \rho_{SL} C_{L\max} V_{TO}^2}{2\beta k_{TO}^2} \quad (2.45)$$

where the takeoff velocity (V_{TO}) is the sum of the catapult end speed (V_{end}) and the wind-over-deck (V_{wod}) or

$$V_{TO} = V_{end} + V_{wod} \quad (2.46)$$

A typical value of k_{TO} is 1.1 and of V_{end} is 120 kn (nautical miles per hour). Wind-over-deck can be 20 to 40 kn, but design specifications may require launch with zero wind-over-deck or even a negative value to ensure launch at anchor. This constraint boundary is simply a vertical line on a plot of thrust loading vs wing loading with the minimum thrust loading given, as already seen in Eq. (2.43), by

$$\left[\frac{T_{SL}}{W_{TO}} \right]_{\min} = \frac{\beta}{\alpha} \left\{ K_1 \frac{C_{L\max}}{k_{TO}^2} + K_2 + \frac{C_{D0} + C_{DR}}{C_{L\max}/k_{TO}^2} + \frac{1}{g_0} \frac{dV}{dt} \right\} \quad (2.47)$$

where α , K_1 , K_2 , and C_{D0} are evaluated at static conditions. A typical value of the required minimum horizontal acceleration at the end of the catapult (dV/dt) is $0.3 g_0$.

2.2.11 Case 11: Carrier Landing

Given: $n = 1$ ($L = W$), V_{TD} , $C_{L\max}$, k_{TD} , β , and the values of h and σ . Rewriting Eq. (2.45) for the touchdown condition gives

$$\left[\frac{W_{TO}}{S} \right]_{\max} = \frac{\sigma \rho_{SL} C_{L\max} V_{TD}^2}{2\beta k_{TD}^2} \quad (2.48)$$

where the touchdown velocity (V_{TD}) is the sum of the engagement speed (V_{eng} , the speed of the aircraft relative to the carrier) and the wind-over-deck (V_{wod}), or

$$V_{TD} = V_{eng} + V_{wod}$$

A typical value of k_{TD} is 1.15 and of V_{eng} is 140 kn (nautical miles per hour). As in Case 10, this constraint boundary is simply a vertical line on a plot of thrust

loading vs wing loading. The minimum thrust loading is given by Eq. (2.49):

$$\left[\frac{T_{SL}}{W_{TO}} \right]_{\min} = \frac{\beta}{\alpha} \left\{ K_1 \frac{C_{L \max}}{k_{TD}^2} + K_2 + \frac{C_{D0} + C_{DR}}{C_{L \max}/k_{TD}^2} + \sin \theta \right\} \quad (2.49)$$

where $-\theta$ is the glide-slope angle.

2.2.12 Case 12: Carrier Approach (Wave-off)

Given: θ , $n = 1$ ($L = W$), V_{TD} , dV/dt , α , β , C_{DR} , $C_{L \max}$, k_{TD} , and the values of h and σ .

Because carrier pilots do not flair and slow down for landing but fly right into the carrier deck in order to make certain the tail hook catches the landing cable, the approach speed is the same as the touchdown speed (V_{TD}). Rewriting Eq. (2.48) for the approach condition gives

$$\left[\frac{W_{TO}}{S} \right]_{\max} = \frac{\sigma \rho_{SL} C_{L \max} V_{TD}^2}{2\beta k_{TD}^2} \quad (2.50)$$

As in Cases 10 and 11, this constraint boundary is simply a vertical line on a plot of thrust loading vs wing loading.

Under these conditions Eq. (2.11) becomes

$$\left[\frac{T_{SL}}{W_{TO}} \right]_{\min} = \frac{\beta}{\alpha} \left\{ K_1 \frac{C_{L \max}}{k_{TD}^2} + K_2 + \frac{C_{D0} + C_{DR}}{C_{L \max}/k_{TD}^2} + \sin \theta + \frac{1}{g_0} \frac{dV}{dt} \right\} \quad (2.51)$$

where $-\theta$ is the glide-slope angle and α , K_1 , K_2 , and C_{D0} are evaluated at static conditions. Typical wave-off requirements are an acceleration of $1/8 g_0$ while on a glide-slope of 4 deg.

2.3 Preliminary Estimates for Constraint Analysis

Preliminary estimates of the aerodynamic characteristics of the airframe and of the installed engine thrust lapse are required before the constraint analysis can be done. The following material is provided to help obtain these preliminary estimates.

2.3.1 Aerodynamics

The maximum coefficient of lift ($C_{L \max}$) enters into the constraint analysis during the takeoff and landing phases. Typical ranges for $C_{L \max}$ divided by the cosine of the sweep angle at the quarter chord ($\Lambda_{c/4}$) are presented in Table 2.1, which is taken from Ref. 4. This table provides typical $C_{L \max}$ for cargo- and passenger-type aircraft. For fighter-type aircraft a clean wing will have $C_{L \max} \sim 1.0 \rightarrow 1.2$ and a wing with a leading edge slat will have $C_{L \max} \sim 1.2 \rightarrow 1.6$. The takeoff maximum lift coefficient is typically 80% of the landing value.

Table 2.1 $C_{L\max}$ for high lift devices⁴

High lift device		Typical flap angle, deg		$C_{L\max}/\cos(\Lambda_{c/4})$	
Trailing edge	Leading edge	Takeoff	Landing	Takeoff	Landing
Plain	—	20	60	1.4 → 1.6	1.7 → 2.0
Single slot	—	20	40	1.5 → 1.7	1.8 → 2.2
Fowler	—	15	40	2.0 → 2.2	2.5 → 2.9
Double sltd.	—	20	50	1.7 → 2.0	2.3 → 2.7
Double sltd.	Slat	20	50	2.3 → 2.6	2.8 → 3.2
Triple sltd.	Slat	20	40	2.4 → 2.7	3.2 → 3.5

The lift-drag polar for most large cargo and passenger aircraft can be estimated (Ref. 5) by using Fig. 2.9 and Eq. (2.9) with

$$0.001 \leq K'' \leq 0.03, \quad 0.1 \leq C_{L\min} \leq 0.3, \quad K' = \frac{1}{\pi A Re}$$

where the wing aspect ratio (AR) is between 7 and 10 and the wing planform efficiency factor (e) is between 0.75 and 0.85. Note that

$$K_1 = K' + K'', \quad C_{D0} = C_{D\min} + K''C_{L\min}^2, \quad K_2 = -2K''C_{L\min}$$

The lift-drag polar for high-performance fighter-type aircraft can be estimated (Ref. 5) using Eq. (2.9), $K_2 = 0$, and Figs. 2.10 and 2.11.

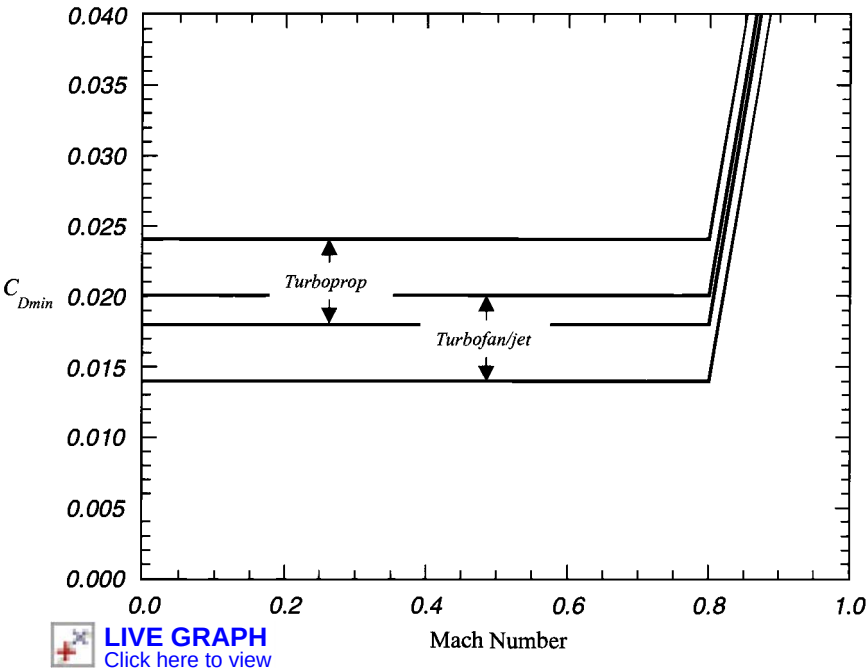


Fig. 2.9 $C_{D\min}$ for cargo and passenger aircraft.⁵

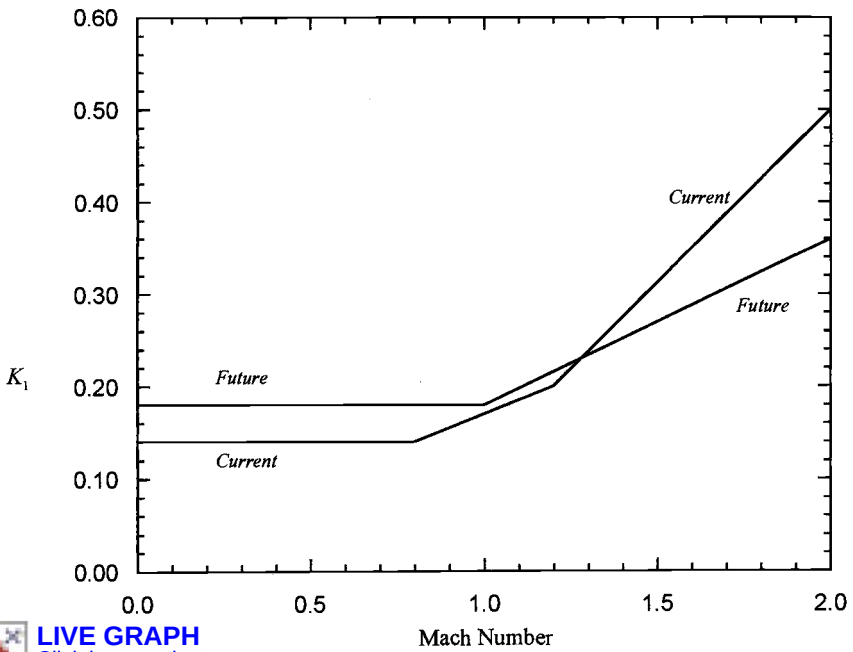


Fig. 2.10 K_1 for fighter aircraft.⁵

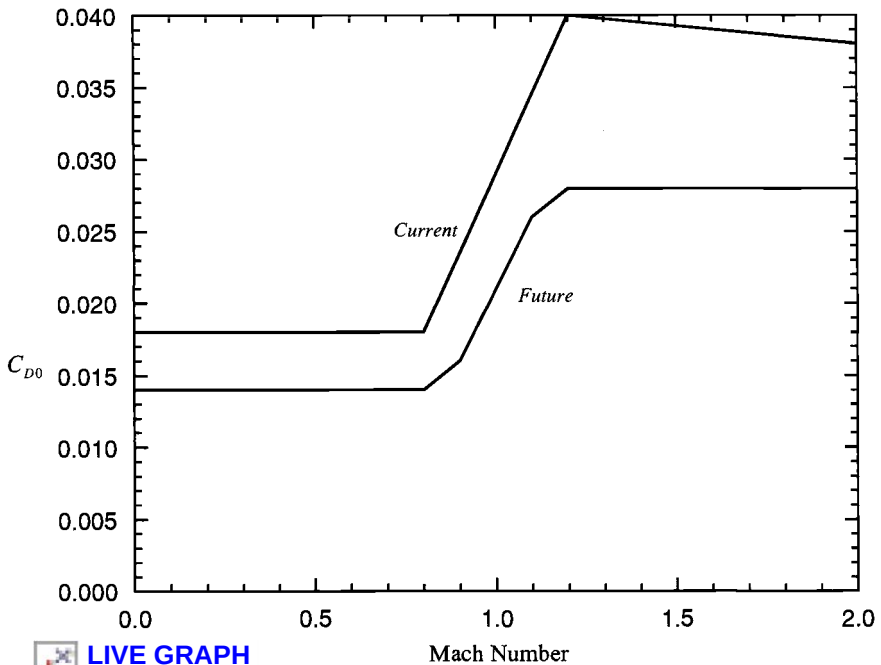


Fig. 2.11 C_{D0} for fighter aircraft.⁵

2.3.2 Propulsion

The variation of installed engine thrust with Mach number, altitude, and afterburner operation can be estimated by developing a simple algebraic equation that has been fit to either existing data of company-published performance curves or predicted data based on the output of performance cycle analysis (see Chapter 5) with estimates made for installation losses. The following algebraic equations for installed engine thrust lapse are based on the expected performance of advanced engines in the 2000 era and beyond. These models use the nondimensional temperature (θ and θ_0) and nondimensional pressure (δ and δ_0), defined next, and the throttle ratio (TR) described in Appendix D. The rationale for these models has a sound, fundamental basis that is developed in Mattingly⁶ and confirmed by the analysis of Appendix D.

$$\theta = T/T_{std} \quad \text{and} \quad \theta_0 = T_t/T_{std} = \theta \left(1 + \frac{\gamma - 1}{2} M_0^2 \right) \quad (2.52a)$$

$$\delta = P/P_{std} \quad \text{and} \quad \delta_0 = P_t/P_{std} = \delta \left(1 + \frac{\gamma - 1}{2} M_0^2 \right)^{\frac{\gamma}{\gamma - 1}} \quad (2.52b)$$

High bypass ratio turbofan ($M_0 < 0.9$)

$$\begin{aligned} \theta_0 \leq TR \quad \alpha &= \delta_0 \{ 1 - 0.49\sqrt{M_0} \} \\ \theta_0 > TR \quad \alpha &= \delta_0 \left\{ 1 - 0.49\sqrt{M_0} - \frac{3(\theta_0 - TR)}{1.5 + M_0} \right\} \end{aligned} \quad (2.53)$$

Low bypass ratio, mixed flow turbofan

Maximum power:

$$\begin{aligned} \theta_0 \leq TR \quad \alpha &= \delta_0 \\ \theta_0 > TR \quad \alpha &= \delta_0 \{ 1 - 3.5(\theta_0 - TR)/\theta_0 \} \end{aligned} \quad (2.54a)$$

Military power:

$$\begin{aligned} \theta_0 \leq TR \quad \alpha &= 0.6\delta_0 \\ \theta_0 > TR \quad \alpha &= 0.6\delta_0 \{ 1 - 3.8(\theta_0 - TR)/\theta_0 \} \end{aligned} \quad (2.54b)$$

Turbojet

Maximum power:

$$\begin{aligned} \theta_0 \leq TR \quad \alpha &= \delta_0 \{ 1 - 0.3(\theta_0 - 1) - 0.1\sqrt{M_0} \} \\ \theta_0 > TR \quad \alpha &= \delta_0 \left\{ 1 - 0.3(\theta_0 - 1) - 0.1\sqrt{M_0} - \frac{1.5(\theta_0 - TR)}{\theta_0} \right\} \end{aligned} \quad (2.55a)$$

Military power:

$$\begin{aligned} \theta_0 \leq TR \quad \alpha &= 0.8\delta_0 \{ 1 - 0.16\sqrt{M_0} \} \\ \theta_0 > TR \quad \alpha &= 0.8\delta_0 \left\{ 1 - 0.16\sqrt{M_0} - \frac{24(\theta_0 - TR)}{(9 + M_0)\theta_0} \right\} \end{aligned} \quad (2.55b)$$

Turboprop

$$\begin{aligned}
 M_0 &\leq 0.1 & \alpha &= \delta_0 \\
 \theta_0 &\leq TR & \alpha &= \delta_0 \{1 - 0.96(M_0 - 1)^{1/4}\} \\
 \theta_0 &> TR & \alpha &= \delta_0 \left\{ 1 - 0.96(M_0 - 1)^{1/4} - \frac{3(\theta_0 - TR)}{8.13(M_0 - 0.1)} \right\}
 \end{aligned} \quad (2.56)$$

2.3.3 Weight Fraction

In the computation of every constraint, the instantaneous weight fraction (β) of the aircraft is required. An initial numerical estimate for this weight fraction at each mission junction must be based on experience. To provide you with guidance for these estimates, the instantaneous weight fraction along the mission of two typical aircraft, a fighter aircraft and a cargo or passenger aircraft, are shown in Figs. 2.12 and 2.13, respectively.

2.4 Example Constraint Analysis

The performance requirements of the Air-to-Air Fighter (AAF) aircraft Request for Proposal (RFP) in Chapter 1 that might constrain the permissible aircraft takeoff thrust loading (T_{SL}/W_{TO}) and wing loading (W_{TO}/S) are listed in Table 2.E1. In this example, we shall translate each of these performance requirements into a constraint boundary on a diagram of takeoff thrust loading vs wing loading. This will be accomplished with each requirement by developing an equation for the limiting allowable loadings in the form

$$f\{(T_{SL}/W_{TO}), (W_{TO}/S)\} = 0$$

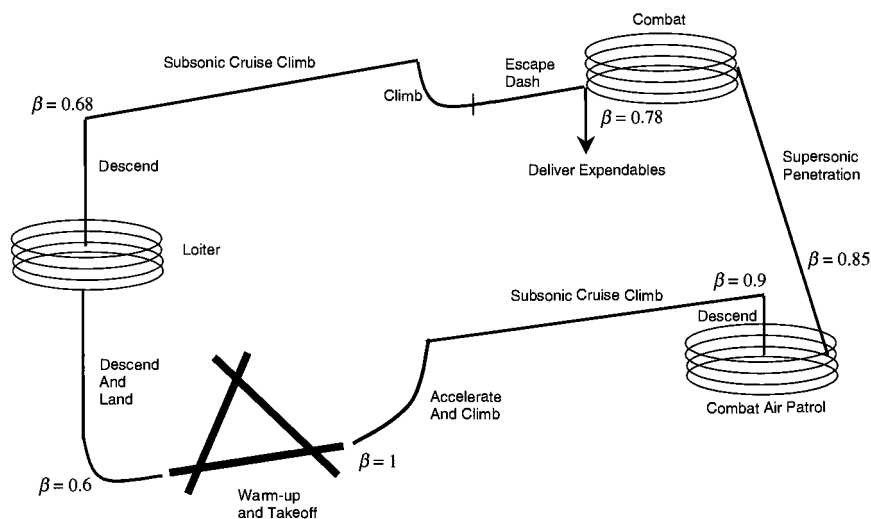


Fig. 2.12 Instantaneous weight fraction: typical fighter aircraft.

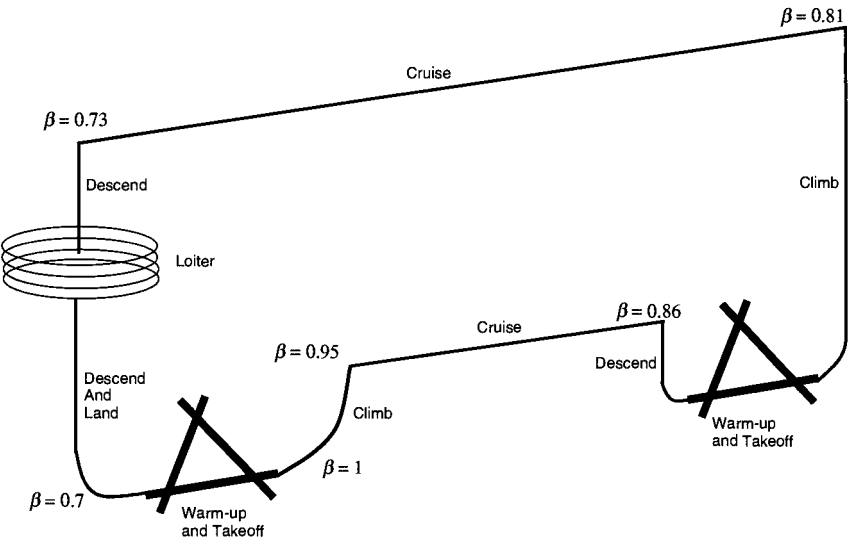


Fig. 2.13 Instantaneous weight fraction—typical cargo and passenger aircraft.

Finally, with the help of the constraint diagram so constructed, we shall select initial values of (T_{SL}/W_{TO}) and (W_{TO}/S) for the AAF.

2.4.1 Constraint Boundary Analysis

To proceed, it is necessary to have preliminary estimates for $C_{L\max}$, the lift-drag polar, and engine performance data. These data are based on current technology

Table 2.E1 Selected AAF specifications

Mission phases and segments		Performance requirement
1–2	Takeoff	2000 ft PA, 100°F, $s_{TO} = s_G + s_R \leq 1500$ ft
	Acceleration	$k_{TO} = 1.2, \mu_{TO} = 0.05$, max power
	Rotation	$V_{TO}, t_R = 3$ s, max power
6–7 and 8–9	Supersonic penetration and escape dash	1.5M/30 kft, no afterburning (if possible)
7–8	Combat	30,000 ft
	Turn 1	1.6M, one 360 deg 5g sustained turn, with afterburning
	Turn 2	0.9M, two 360 deg 5g sustained turn, with afterburning
	Acceleration	$0.8 \rightarrow 1.6$ M, $\Delta t \leq 50$ s, max power
13–14	Landing	2000 ft PA, 100°F, $s_L = s_{FR} + s_{BR} \leq 1500$ ft
	Free roll	$k_{TD} = 1.15, t_{FR} = 3$ s, $\mu_B = 0.18$
	Braking	Drag chute diameter 15.6 ft, deployment ≤ 2.5 s
Max Mach number RFP para. C.1		1.8M/40 kft, max power

for fighter-type aircraft and engines. The variation of thrust with Mach number and density is shown in Fig. 2.E1b that depicts the so-called engine thrust lapse of Eqs. (2.54a) and (2.54b) for three typical values of the throttle ratio (TR). In addition, the instantaneous weight fraction (β) must be estimated for each item of Table 2.E1. Reasonable estimates (see Figs. 2.12 and 2.13) will fall between 1.0 at takeoff and 0.5, say, at landing.

The aerodynamic data of Fig. 2.E1a provide the initial estimates of $C_{L\max}$, K_1 , and C_{D0} required by Eq. (2.11b) and its descendants. Please note that the AAF is at first assumed to have an uncambered airfoil, for which $K_2 = 0$. The following AAF calculations will therefore use the relevant equations with K_2 set equal to zero. Note that $C_{L\max} = 2$ was arrived at after many values of $C_{L\max}$ were examined, such that takeoff would not overconstrain the problem. According to Table 2.1, this requires a fighter with very low wing sweep (improbable, especially in this case with supercruise requirements) or augmented high lift (such as achieved by using vortex lift created by leading edge extensions, etc.). This challenges the aircraft designer to join in the effort by using advanced technology for this requirement.

The installed propulsion data of a low bypass ratio, mixed flow turbofan engine are selected and plotted in Fig. 2.E1b based on Eqs. (2.54a) and (2.54b). A range of TR from 1.0 to 1.08 is shown because the hot day takeoff and supercruise requirements desire a TR greater than 1.0 (see Appendix D). It is important to have a good first estimate of the installed engine thrust lapse, which is usually obtained from such open literature information as company brochures, technical papers, and textbooks.^{1,2} Another very effective method for obtaining engine thrust lapse is simply to run the performance computer portion of the AEDsys program supplied with this textbook for an initial guess of the engine design point. That was the origin of the engine thrust lapse equations of Sec. 2.3.2.

The computations and data required to construct each boundary in the complete constraint diagram of the AAF are contained in the Sec. 2.4.3. The Constraint Analysis software in the AEDsys program makes these calculations effortless.

Before determining the constraint for each item of Table 2.E1 in detail, we will first illustrate the procedure with the takeoff distance and supercruise phases. This will also help us appreciate the role of the engine TR and select a value capable of meeting these two constraints.

Takeoff. For this illustrative case we consider the airplane accelerated by thrust with no resisting forces whatever in the ground roll. The thrust is balanced by drag forces during the constant velocity rotation. Under these conditions our takeoff constraint boundary equation comes from Case 5, Eq. (2.22), solved for s_G , and Eq. (2.26), where $s_{TO} = s_G + s_R$, or

$$s_{TO} = \left\{ \frac{k_{TO}^2 \beta^2}{\rho g_0 C_{L\max} \alpha_{wet} (T_{SL}/W_{TO})} \right\} \left(\frac{W_{TO}}{S} \right) + \left\{ t_R k_{TO} \sqrt{\frac{2\beta}{\rho C_{L\max}}} \right\} \sqrt{\frac{W_{TO}}{S}} \quad (2.E1)$$

which is in the form

$$a \left(\frac{W_{TO}}{S} \right) + b \sqrt{\frac{W_{TO}}{S}} - c = 0$$

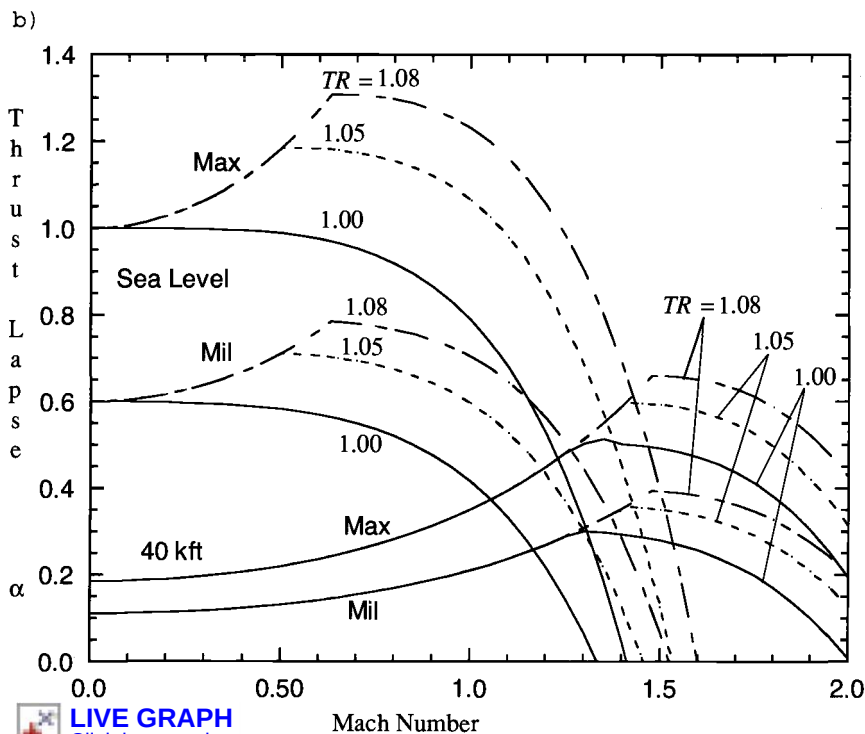
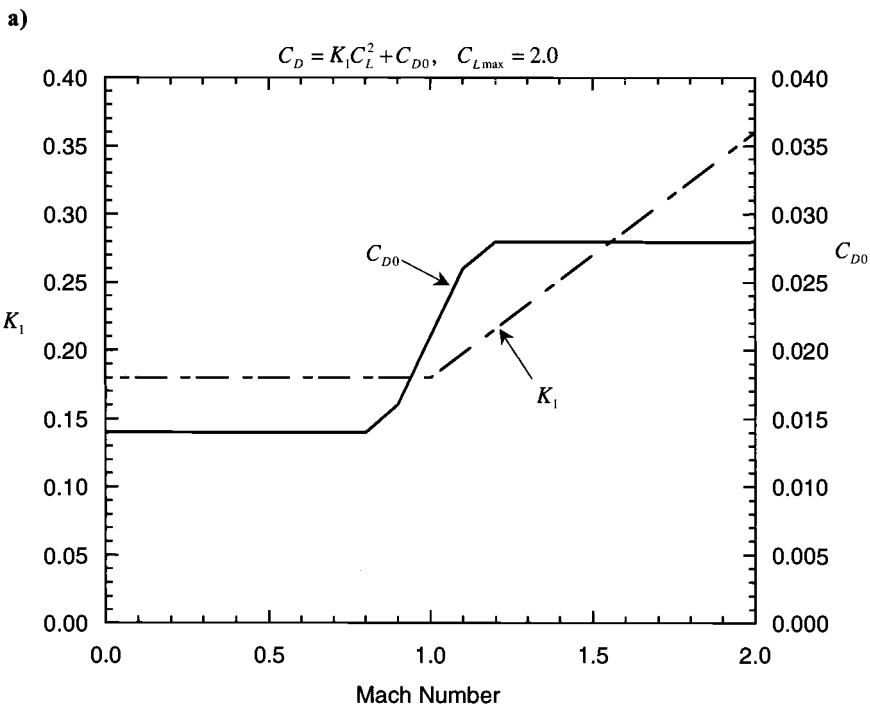


Fig. 2.E1 Preliminary AAF data: a) aerodynamic data; b) installed thrust lapse.

Table 2.E2 Thrust lapse variation with throttle ratio-AB on

TR	α_{wet}
1.00	0.6885
1.05	0.8400
1.06	0.8703
1.07	0.9006
1.08	0.9309

From item 1 of Table 2.E1, Fig. 2.E1, and the standard atmosphere of Appendix B, we have

$$\begin{aligned}
 k_{TO} &= 1.2 & \beta &= 1.0 & \theta &= 1.0796 & (\theta_0)_{M=0.1} &= 1.0818 \\
 M_{TO} &= 0.1 & C_{L\max} &= 2.0 & \delta &= 0.9298 & (\delta_0)_{M=0.1} &= 0.9363 \\
 \sigma &= \delta/\theta = 0.8612 & \rho &= 0.002047 \text{ sl/ft}^3 & t_R &= 3.0 \text{ s} & s_{TO} &= 1500 \text{ ft} \\
 g_0 &= 32.17 \text{ ft/s}^2
 \end{aligned}$$

For the range of TR the corresponding thrust lapses (α) from Eq. (2.54a) are shown in Table 2.E2.

With these values we can evaluate the coefficients a , b , and c in Eq. (2.E1), which, for computational convenience, we solve for W_{TO}/S in order to obtain

$$\left(\frac{W_{TO}}{S} \right) = \left\{ \frac{-b + \sqrt{b^2 + 4ac}}{2a} \right\}^2 \quad (2.E2)$$

where

$$a = \frac{10.93}{\alpha_{wet}(T_{SL}/W_{TO})} \quad b = 79.57 \quad c = 1500$$

whence we obtain the matrix of results for W_{TO}/S :

TR	T_{SL}/W_{TO}	0.4	0.8	1.2	1.6	2.0	2.4
1.00	W_{TO}/S (lbf/ft ²)	27.3	47.9	64.9	79.6	92.5	104
1.05	W_{TO}/S (lbf/ft ²)	32.2	55.7	74.8	91.1	105	118
1.06	W_{TO}/S (lbf/ft ²)	33.2	57.2	76.7	93.2	107	120
1.07	W_{TO}/S (lbf/ft ²)	34.1	58.7	78.6	95.33	110	122
1.08	W_{TO}/S (lbf/ft ²)	35.1	60.2	80.4	97.4	112	125

The takeoff constraint boundaries for this range of throttle ratios are shown in Fig. 2.E2. An airplane with thrust and wing loadings in the region to the left of the boundary can take off within the specified constraints, but those to the right cannot.

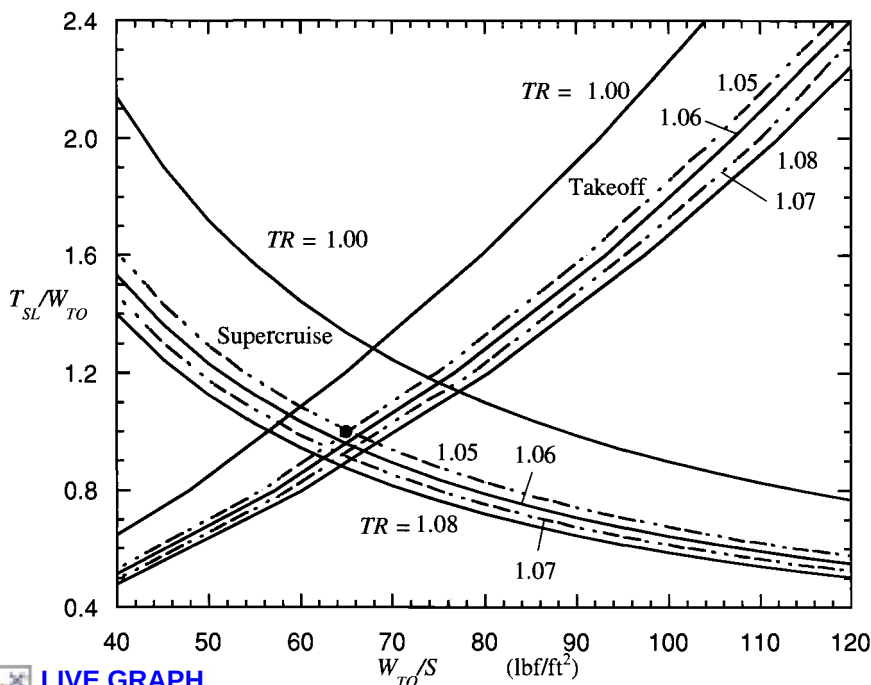


Fig. 2.E2 Constraint diagram for takeoff and supercruise conditions.

Supercruise. For constant altitude/speed cruise, Case 1, Eq. (2.12), with $C_{DR} = K_2 = 0$ gives

$$\left(\frac{T_{SL}}{W_{TO}} \right) = \frac{\beta}{\alpha} \left\{ K_1 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + \frac{C_{D0}}{\beta/q (W_{TO}/S)} \right\}$$

From item 2 of Table 2.E1, Fig. 2.E1, and the standard atmosphere of Appendix B, we have

$$\begin{aligned} \theta &= 0.7940 & (\theta_0)_{M=1.5} &= 1.1513 & \delta &= 0.2975 & (\delta_0)_{M=1.5} &= 1.0921 \\ K_1 &= 0.27 & C_{D0} &= 0.028 & \beta &= 0.78 \\ q &= 991.6 \text{ lb/ft}^2 \end{aligned}$$

The supercruise constraint boundary equation is, therefore,

$$\left(\frac{T_{SL}}{W_{TO}} \right) = \frac{0.78}{\alpha} \left\{ 2.12 \times 10^{-4} \left(\frac{W_{TO}}{S} \right) + \frac{35.6}{(W_{TO}/S)} \right\} \quad (2.E3)$$

**Table 2.E3 Thrust lapse variation
with throttle ratio-AB off**

TR	α_{dry}
1.00	0.3278
1.05	0.4359
1.06	0.4576
1.07	0.4792
1.08	0.5008

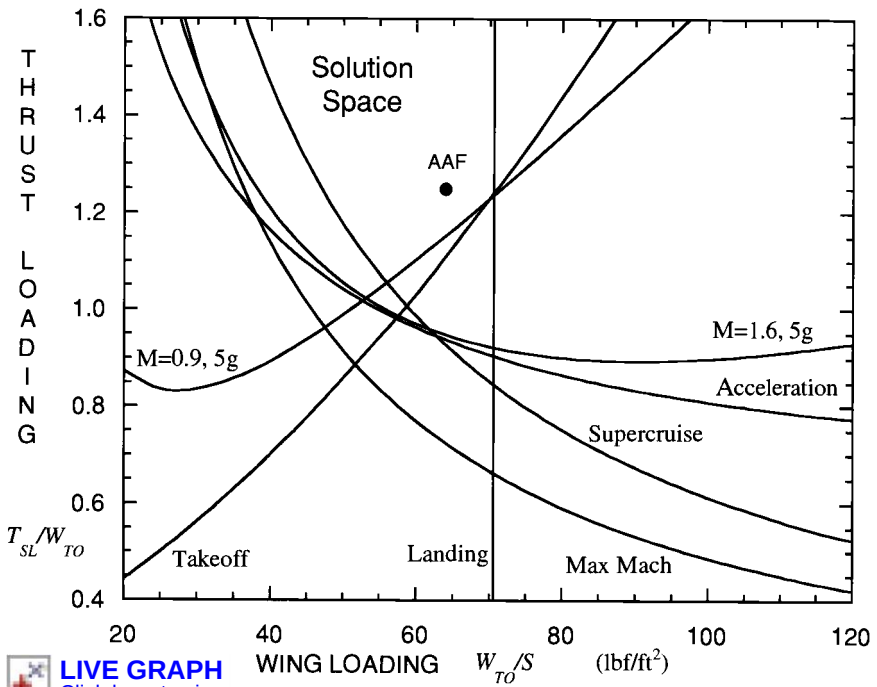
with $\alpha = \alpha_{dry}$ as a function of TR from Eq. (2.54b), as shown in Table 2.E3, whence we obtain the matrix of results for T_{SL}/W_{TO} :

TR	W_{TO}/S (lbf/ft ²)	20	40	60	80	100	120
1.00	T_{SL}/W_{TO}	4.25	2.14	1.44	1.10	0.898	0.767
1.05	T_{SL}/W_{TO}	3.19	1.61	1.08	0.827	0.675	0.576
1.06	T_{SL}/W_{TO}	3.04	1.53	1.03	0.788	0.643	0.549
1.07	T_{SL}/W_{TO}	2.90	1.46	0.986	0.752	0.614	0.524
1.08	T_{SL}/W_{TO}	2.80	1.40	0.944	0.720	0.588	0.502

As shown in Fig. 2.E2, this constraint boundary places a lower limit on the allowable wing loading and, together with the takeoff constraint boundary, encloses the “solution space” of allowable combinations of thrust and wing loadings that satisfy the two performance requirements considered here. The main consequence of increased throttle ratio, as expected, is to reduce T_{SL}/W_{TO} by sustaining thrust to higher values of M_0 and θ_0 (see Fig. 2.E1b).

The selection of a thrust loading and wing loading from Fig. 2.E2 is a compromise of many factors. For a given W_{TO} , a low (W_{TO}/S) value means large wing area while a high value of (T_{SL}/W_{TO}) results in a large thrust requirement. In addition, a low wing loading reduces the airplane riding qualities and range and can increase the aircraft radar cross section. We would prefer, therefore, relatively low thrust and high wing loadings. We might, based on the constraint diagram of Fig. 2.E2, select 1.0 and 64 lbf/ft² for our AAF thrust and wing loadings and an engine throttle ratio of 1.07. However, we must take into account all performance requirements as well as the takeoff ground roll drag in constructing the fighter’s complete constraint diagram.

When this is done, we employ the AEDsys Constraint Analysis software to obtain and construct Fig. 2.E3, which is the complete constraint diagram we seek in order to make a judicious selection of (T_{SL}/W_{TO}) and (W_{TO}/S) for the AAF. Notice that the solution space in the diagram is bounded with constraints formed by the supercruise, 0.9M/5g combat turn, takeoff, and landing RFP requirements. It is interesting to superimpose the AAF solution space of Fig. 2.E3 on Fig. 2.3, which contains the wing and thrust loadings of 22 fighter-type airplanes, as is done in the composite Fig. 2.E4. We see that none of the contemporary fighters meets all of the RFP requirements for the AAF, thus the need for a new airplane. However, the larger YF-22 and YF-23 have similar thrust and wing loadings, which might be expected because they have similar requirements.



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Fig. 2.E3 The complete preliminary AAF constraint diagram ($TR = 1.07$).

2.4.2 Selection of the Preliminary Air-to-Air Fighter Design Point

We are now in a position to make a preliminary selection of the thrust loading and wing loading that will guarantee that all the flight constraints are met. To do this, we will refer repeatedly to the constraint diagram of Fig. 2.E3. You will find that the tools are provided to allow the solution to be iterated and refined as better information becomes available, until a final converged solution is obtained.

As you can see from Fig. 2.E3, the limits imposed by the sustained 1.6M/5g combat turn, the acceleration requirement, and the maximum Mach are, for this aircraft, not important. Please bear in mind, however, that this will not be the case for aircraft designed for other applications and may not even remain true for the AAF if the underlying assumptions change (e.g., if the afterburner is not used during sustained combat turns). For the time being, then, we will concentrate on the four flight conditions that do form the boundaries of the AAF solution space, namely, supercruise, 0.9M/5g combat turn, takeoff, and landing.

At this stage of the design process, the constraint analysis also can be used to bring down the ultimate takeoff weight (and cost) of the AAF. On the one hand, lower thrust loadings lead to smaller engines and greater wing loadings lead to smaller wings. On the other hand, as you will see in Sec. 3.2.7, the fuel consumed in supercruise can be reduced by selecting a wing loading closer to the minimum thrust loading, which is evidently to the right of the solution space. This means that we must focus on design points on the right side of the “bucket” formed

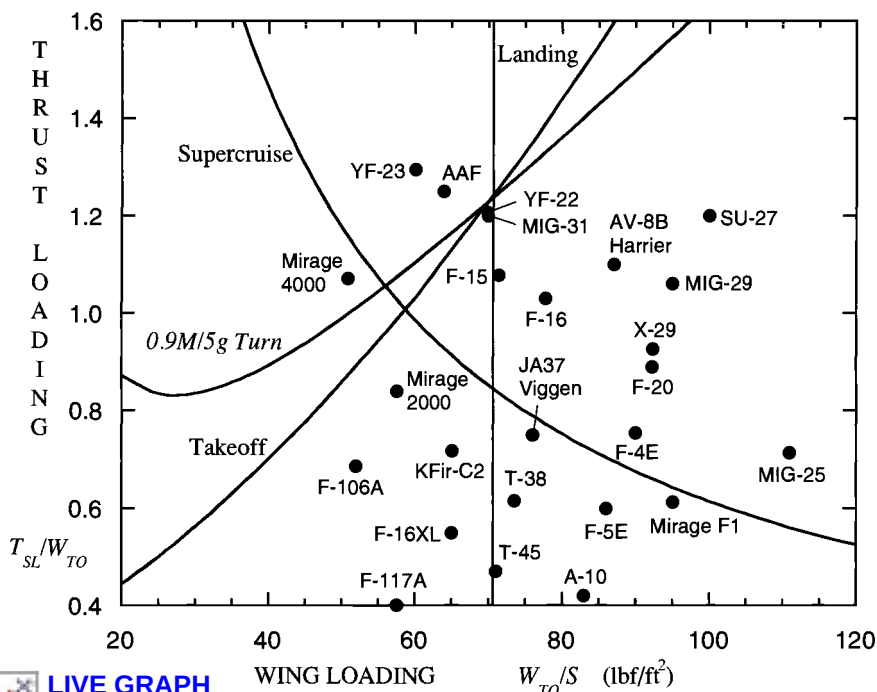


Fig. 2.E4 Composite thrust loading vs wing loading—fighter aircraft and preliminary AAF.

near the juncture of the three constraining lines, preferably those near to the region of previous thrust loading experience, which extends to about $T_{SL}/W_{TO} = 1.2$ according to Fig. 2.E4. Fortunately, the wing loadings in the region are well within the design experience or the industry, as seen in Fig. 2.E4, and high enough to assure good handling qualities.³ It is equally important to avoid trying to put too fine a point on our choice, lest the movement of any of the constraining lines caused by improved estimates of aircraft and/or engine performance render it useless. Following this logic, it appears that the preliminary design point

$$T_{SL}/W_{TO} = 1.25$$

$$W_{TO}/S = 64 \text{ lb/ft}^2$$

$$TR = 1.07$$

as indicated on Figs. 2.E3 and 2.E4, is a reasonable first choice, and this shall be used as our decision. A host of interesting observations evolve, the most important of which are the following:

- 1) The “new” capabilities of the AAF, namely short takeoff and landing, and nonafterburning supercruise have “driven” the design.
- 2) Relieving the takeoff constraint (by increasing s_{TO} or $C_{L\max}$), the landing constraint (by increasing s_L or providing reverse thrust), the 0.9M combat turn (by

decreasing the number of *gs*), and the supercruise constraint (by decreasing the supercruise Mach number) simultaneously would do little to reduce the required T_{SL}/W_{TO} because of the bunching of the other constraints in the same neighborhood. The good side of this situation is that the resulting engine will be running near its full thrust at many flight conditions, so that its size will not be made excessively large by any single requirement, and its performance is liable to remain high over the entire mission. This often happens when the RFP is the result of thoughtful consideration and balancing of the requirements.

3) The drag chute is a bargain. Without it T_{SL}/W_{TO} would increase beyond any reasonable limit because $W_{TO}/S = 50.8 \text{ lbf/ft}^2$ with no drag chute.

4) The throttle ratio has migrated in the direction of the supercruise $\theta_0 = 1.15$ (see Appendix D).

Evaluating P_s . At this point in the design process, it may surprise you to find that a great deal of preliminary information is already available about the AAF. For example, once T_{SL}/W_{TO} and W_{TO}/S are selected, it is possible to compute the weight specific excess power (P_s) for flight at any chosen β , n , altitude, and velocity. To see this, combine Eqs. (2.2b) and (2.11) with $R = 0$ to find that

$$P_s = V \left\{ \frac{\alpha}{\beta} \left(\frac{T_{SL}}{W_{TO}} \right) - K_1 n^2 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) - K_2 n - \frac{C_{D0}}{\beta/q(W_{TO}/S)} \right\} \quad (2.E4)$$

Equation (2.E4) was used in AEDsys to compute the contours of constant P_s for the preliminary AAF of Fig. 2.E3 at military power with $n = 1$ and $\beta = 0.97$. The results, which are displayed in Fig. 2.E5, are remarkably similar to that of “real” airplanes and contain useful guidance for future pilots of the AAF, as well

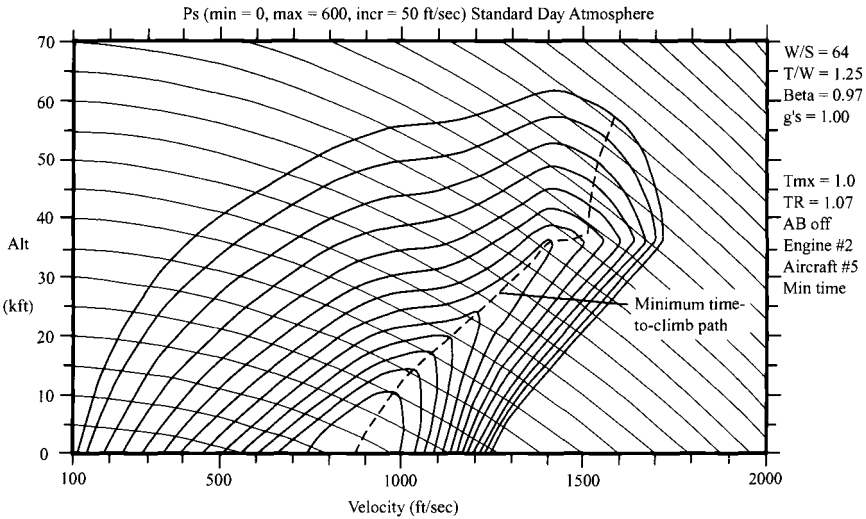


Fig. 2.E5 Preliminary AAF P_s contours at military power with minimum time-to-climb path.



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as for analyses. Fighter pilots routinely compare these P_s diagrams for their aircraft with those of their potential adversaries in order to determine where in the flight envelope they enjoy the greatest combat advantage.

Further, Fig. 2.E5 provides a graphic method for determining the minimum time-to-climb flight path from a lower to a higher energy height. Equation (2.2b) rewritten as

$$\Delta t = \int_1^2 dt = \int_{z_{e1}}^{z_{e2}} \frac{dz_e}{P_s} \quad (2.E5)$$

shows that the minimum time-to-climb from $z_{e1}(h_1, V_1)$ to the higher energy level $z_{e2}(h_2, V_2)$ corresponds to the flight path that produces the maximum specific excess power P_s at each z_e . This maximum occurs at the point of tangency between a line of constant energy height and the contour line of the maximum P_s attainable for that energy height. For $z_e = 70$ kft, for example, this tangency occurs at $P_s = 320$ ft/s, $V = 1480$ ft/s, and $h = 36$ kft. Continuing in this vein, one can find the minimum time-to-climb flight path, from z_e of 13 kft at sea level to a z_e of 96 kft at an altitude of 57 kft, as shown in Fig. 2.E5.

These preliminary quantitative results will, of course, change once the final selection of the AAF design point has been made and as improved models of the aircraft and engine performance are developed and used.

There are several other features of the P_s contours of Fig. 2.E5 that are striking and noteworthy, and they can be usefully generalized to applications beyond the AAF. First, the theta break line of 1.07 at the peak of each contour is clearly visible, and the value of moving it nearer to the supercruise flight condition is evident. Second, the dip of each contour caused by the transonic drag rise near $M = 1$ is also clearly visible. Third, the discontinuity on the right-hand side of each contour marks the atmospheric tropopause. Finally, the minimum time-to-climb path is quite similar to the theta break line because of the dominant influence of engine thrust on P_s , but not identical because aircraft drag also contributes to the calculation [see Eq. (2.E4)].

2.4.3 Complete AAF Constraint Boundary Computations

The constraint computational procedure and results for the performance requirements in Table 2.E1 follow. The same procedure is used in each constraint boundary calculation. First, the applicable constraint boundary equation from Sec. 2.2 is written. Next, data to determine the values of all constants as they appear from left to right in the equation are listed. This is followed by the specific constraint boundary equation resulting from the given data. Finally, some solution sets of this equation are tabulated. These are the data plotted in Figs. 2.E3 and 2.E4. The same calculations can be performed and the boundaries plotted much more rapidly using the Constraint Analysis portion of the AEDsys software included with this textbook. An additional benefit of the software is that it constitutes a checklist of the quantities required for the calculation of any chosen constraint boundary. The individual values were obtained as described in Sec. 2.4.1, but it must be emphasized that they can be altered for the purpose of exploration at the discretion of the reader.

Careful note should be taken of the fact that these hand calculations completely duplicate the AEDsys Constraint Analysis computations obtained by using the AAFAED file. Any minor differences are simply the result of the cumulative effects of rounding off. Thus we see that one of the motivating forces for carrying four significant figures is to minimize the impact of these small imperfections on the overall results.

Mission phase 1-2: Takeoff, no obstacle. 2000 ft PA, 100°F, $k_{TO} = 1.2$, $\mu_{TO} = 0.05$, $t_R = 3$ s, $s_{TO} = s_G + s_R \leq 1500$ ft, max power.

From Case 6, Eqs. (2.25) and (2.26) with $s_{TO} = s_G + s_R$

$$\left(\frac{W_{TO}}{S}\right) = \left\{ -b + \sqrt{\frac{b^2 + 4ac}{2a}} \right\}^2$$

where

$$a = -\frac{\beta}{\rho g_0 \xi_{TO}} \ln \left\{ 1 - \xi_{TO} \left/ \left[\left(\frac{\alpha}{\beta} \frac{T_{SL}}{W_{TO}} - \mu_{TO} \right) \frac{C_{L \max}}{k_{TO}^2} \right] \right\} \right\}$$

$$b = t_R k_{TO} \sqrt{2\beta/(\rho C_{L \max})} \quad c = s_{TO}$$

We estimate α and β to be constant at appropriate mean values. A conservative estimate for ξ_{TO} for the AAF is obtained by assuming $(C_{DR} - \mu_{TO} C_L) = 0$ and evaluating C_D at $C_L = C_{L \max}/k_{TO}^2$. With

$\beta = 1.0$	$\rho = 0.002047$ slug/ft ³	$C_{L \max} = 2.0$	$C_L = 1.389$
$K_1 = 0.18$	$C_{D0} = 0.014$	$\xi_{TO} = 0.3612$	
$\sigma = 0.8613$	$\theta = 1.0796$	$\delta = 0.9298$	$TR = 1.07$
$M_{TO} = 0.1$	$(\theta_0)_{M=0.1} = 1.0818$	$(\delta_0)_{M=0.1} = 0.9363$	$(a_{wet})_{M=0.1} = 0.9006$
$s_{TO} = 1500$ ft			

then

$$a = -42.06 \ln \left\{ 1 - 0.2599 \left/ \left[0.9006 \left(\frac{T_{SL}}{W_{TO}} \right) - 0.05 \right] \right\} \right\} \quad b = 79.57 \quad c = 1500$$

whence

T_{SL}/W_{TO}	W_{TO}/S , lbf/ft ²
0.4	15.5
0.8	46.4
1.2	68.7
1.6	86.9
2.0	103

Mission phases 6–7 and 8–9: Supersonic penetration and escape dash.
1.5M/30 kft, no afterburning.

From Case 1, Eq. (2.12), with C_{DR} and $K_2 = 0$

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + \frac{C_{D0}}{\beta/q(W_{TO}/S)} \right\}$$

With

$$\begin{aligned} \beta &= 0.78 & \theta &= 0.7940 & \theta_0 &= 1.1513 & \delta &= 0.2975 \\ \delta_0 &= 1.0921 & TR &= 1.07 & \alpha_{dry} &= 0.4792 & q &= 991.6 \text{ lbf/ft}^2 \\ K_1 &= 0.27 & C_{D0} &= 0.028 \end{aligned}$$

then

$$\left(\frac{T_{SL}}{W_{TO}} \right) = 3.457 \times 10^{-4} \left(\frac{W_{TO}}{S} \right) + \frac{57.94}{(W_{TO}/S)}$$

whence

W_{TO}/S , lbf/ft ²	T_{SL}/W_{TO}
20	2.90
40	1.46
60	0.986
80	0.752
100	0.614
120	0.524

Mission phase 7–8: Combat turn 1. 1.6M/30 kft, one 360 deg 5g sustained turn, with afterburning.

From Case 3, Eq. (2.15), with C_{DR} and $K_2 = 0$

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 n^2 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + \frac{C_{D0}}{\beta/q(W_{TO}/S)} \right\}$$

With

$$\begin{aligned} \beta &= 0.78 & \theta &= 0.7940 & \theta_0 &= 1.2005 & \delta &= 0.2975 \\ \delta_0 &= 1.2645 & TR &= 1.07 & \alpha_{wet} &= 0.7829 & q &= 1128 \text{ lbf/ft}^2 \\ C_{D0} &= 0.028 & K_1 &= 0.288 \end{aligned}$$

then

$$\left(\frac{T_{SL}}{W_{TO}} \right) = 4.960 \times 10^{-3} \left(\frac{W_{TO}}{S} \right) + \frac{40.34}{(W_{TO}/S)}$$

whence

W_{TO}/S , lbf/ft ²	T_{SL}/W_{TO}
20	2.12
40	1.21
60	0.970
80	0.901
100	0.899
120	0.931

Mission phase 7-8: Combat turn 2. 0.9M/30 kft, two 360 deg 5g sustained turns, with afterburning.

From Case 3, Eq. (2.15), with C_{DR} and $K_2 = 0$

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 n^2 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + \frac{C_{D0}}{\beta/q(W_{TO}/S)} \right\}$$

With

$$\begin{aligned} \beta &= 0.78 & \theta &= 0.7940 & \theta_0 &= 0.9226 & \delta &= 0.2975 \\ \delta_0 &= 0.5032 & TR &= 1.07 & \alpha_{wet} &= 0.5033 & q &= 357.0 \text{ lbf/ft}^2 \\ C_{D0} &= 0.016 & K_1 &= 0.18 \end{aligned}$$

then

$$\left(\frac{T_{SL}}{W_{TO}} \right) = 0.01524 \left(\frac{W_{TO}}{S} \right) + \frac{11.35}{(W_{TO}/S)}$$

whence

W_{TO}/S , lbf/ft ²	T_{SL}/W_{TO}
20	0.872
40	0.893
60	1.10
80	1.36
100	1.64
120	1.92

Mission phase 7-8: Horizontal acceleration. 0.8 → 1.6M/30 kft, at $\Delta t \leq 50$ s, max power.

From Case 4, Eq. (2.18a), with C_{DR} and $K_2 = 0$

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + \frac{C_{D0}}{\beta/q(W_{TO}/S)} + \frac{a \Delta M}{g_0 \Delta t} \right\}$$

We obtain approximate constant values of q , K_1 , C_{D0} , and α at a mean Mach number of 1.2.

With

$$\begin{aligned}\beta &= 0.78 & \theta &= 0.7940 & \theta_0 &= 1.0227 & \delta &= 0.2975 \\ \delta_0 &= 0.7215 & TR &= 1.07 & \alpha_{wet} &= 0.7216 & q &= 634.6 \text{ lbf/ft}^2 \\ C_{D0} &= 0.028 & K_1 &= 0.216 & a &= 994.8 \text{ ft/s} & \Delta M &= 0.8 \\ \Delta t &= 50 \text{ s}\end{aligned}$$

then

$$\frac{T_{SL}}{W_{TO}} = 2.870 \times 10^{-4} \left(\frac{W_{TO}}{S} \right) + \frac{24.62}{(W_{TO}/S)} + 0.5348$$

whence

$W_{TO}/S, \text{ lbf/ft}^2$	T_{SL}/W_{TO}
20	1.77
40	1.16
60	0.963
80	0.866
100	0.810
120	0.775

Mission phase 13–14: Landing, no reverse thrust. 2000 ft PA, 100°F, $k_{TD} = 1.15$, $t_{FR} = 3 \text{ s}$, $\mu_B = 0.18$, $s_L = s_{FR} + s_B \leq 1500 \text{ ft}$, GFE drag chute, diameter 15.6 ft, deployment $\leq 2.5 \text{ s}$.

From Case 7, Eqs. (2.33) and (2.37) with $s_L = s_{FR} + s_B$

$$\left(\frac{W_{TO}}{S} \right) = \left\{ \frac{-b + \sqrt{b^2 + 4ac}}{2a} \right\}^2$$

where

$$a = \frac{\beta}{\rho g_0 \xi_L} \ln \left\{ 1 + \xi_L / \left[\left(\mu_B + \frac{(-\alpha)}{\beta} \frac{T_{SL}}{W_{TO}} \right) \frac{C_{L\max}}{k_{TD}^2} \right] \right\}$$

$$b = t_{FR} k_{TD} \sqrt{2\beta/(\rho C_{L\max})} \quad c = S_L$$

We assume a drag chute drag coefficient of 1.4 based on the RFP chute area of 191 ft². Estimating the airplane wing area to be about 500 ft², the chute drag coefficient (C_{DRc}) is 0.5348 referred to the wing area. A conservative estimate of ξ_L for the AAF is obtained by evaluating C_D at 0.8 of the touchdown lift coefficient ($C_{L\max}/k_{TD}^2$) and by assuming $(C_{DR} - \mu_B C_L) = 0$.

With

$$\begin{aligned}\beta &= 0.56 & \rho &= 0.002047 \text{ slug/ft}^3 & C_{L\max} &= 2.0 & C_{DRc} &= 0.5348 \\ C_L &= 1.210 & K_1 &= 0.18 & C_{D0} &= 0.014 & C_D &= 0.2775 \\ \xi_L &= 0.8123 & \alpha &= 0 & s_L &= 1500 \text{ ft}\end{aligned}$$

then

$$a = 14.47 \quad b = 57.06 \quad c = 1500$$

whence

$$W_{TO}/S = 70.6 \text{ lbf/ft}^2$$

With no drag chute, the same assumptions and computational methods yield a wing loading constraint of 50.8 lbf/ft².

Maximum Mach number. $M = 1.8$ at 40,000 ft, max power.

From Case 1, Eq. (2.12), with C_{DR} and $K_2 = 0$

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left\{ K_1 \frac{\beta}{q} \left(\frac{W_{TO}}{S} \right) + \frac{C_{D0}}{\beta/q(W_{TO}/S)} \right\}$$

With

$$\beta = 0.78 \quad \theta = 0.7519 \quad (\theta_0)_{M=1.8} = 1.2391 \quad \delta = 0.1858$$

$$(\delta_0)_{M=1.8} = 1.0676 \quad TR = 1.07 \quad \alpha_{wet} = 0.5575 \quad q = 891.8 \text{ lbf/ft}^2$$

$$K_1 = 0.324 \quad C_{D0} = 0.028$$

then

$$\left(\frac{T_{SL}}{W_{TO}} \right) = 3.965 \times 10^{-4} \left(\frac{W_{TO}}{S} \right) + \frac{44.79}{(W_{TO}/S)}$$

whence

$W_{TO}/S, \text{ lbf/ft}^2$	T_{SL}/W_{TO}
20	2.25
40	1.14
60	0.770
80	0.592
100	0.487
120	0.421

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